

AP Physics C: Electricity & Magnetism

The Complete Exam Guide

All 6 units explained with the calculus you need, Gauss's and Ampère's laws, the four free-response question types, and 10 hard practice questions.

Hi, I'm **Pigo!** I'll drop tips all through this guide.

**3 hrs**

85 min multiple choice
+ 95 min free response

42+4

multiple-choice +
free-response questions

6 units

calculus-based
E&M

73–75%

scored 3 or higher
in recent years

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01 THE EXAM AT A GLANCE

Two sections, one score

Your AP score (1–5) combines a multiple-choice section and a free-response section, each worth half. This is a calculus-based course: expect integrals for fields and differential equations for circuits. A calculator and an equation sheet are allowed on the whole exam.

50%

Section I • Multiple choice • 42 questions

50%

Section II • Free response • 4 questions

Exam structure

Section	Weight & timing	What you do
I • Multiple choice	50% of score 1 hr 25 min 42 questions	Single questions and short sets built around a stimulus such as a diagram, graph or data table. Calculator allowed.
II • Free response	50% of score 1 hr 35 min 4 questions	Four question types: Mathematical Routines • Translation Between Representations • Experimental Design and Analysis • Qualitative/Quantitative Translation.

HYBRID DIGITAL

Click in Bluebook, write on paper

You answer multiple-choice questions and read the free-response questions in the **Bluebook** app, then handwrite your free-response answers in a paper booklet. Practice writing clear, organized work by hand.

PROVIDED ON EXAM DAY

Equation sheet + calculator

A table of equations and constants is provided, and a calculator is allowed throughout. Knowing **when** an equation applies, and its conditions, matters more than memorizing it.

How to use this guide

Just starting?

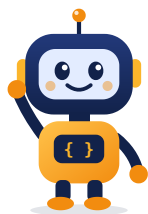
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 22 minutes, then use the final-four-weeks plan.



Pigo's tip: Physics answers are scored on **reasoning**. Start every explanation from a principle, such as Newton's second law or conservation of energy, then apply it to the situation.

SECTION I

02 The multiple-choice section

42 questions in 85 minutes gives you about 2 minutes per question. Many questions test reasoning with symbols and graphs rather than plugging in numbers.

42

questions, some in stimulus-based sets

~2:00

minutes per question on average

50%

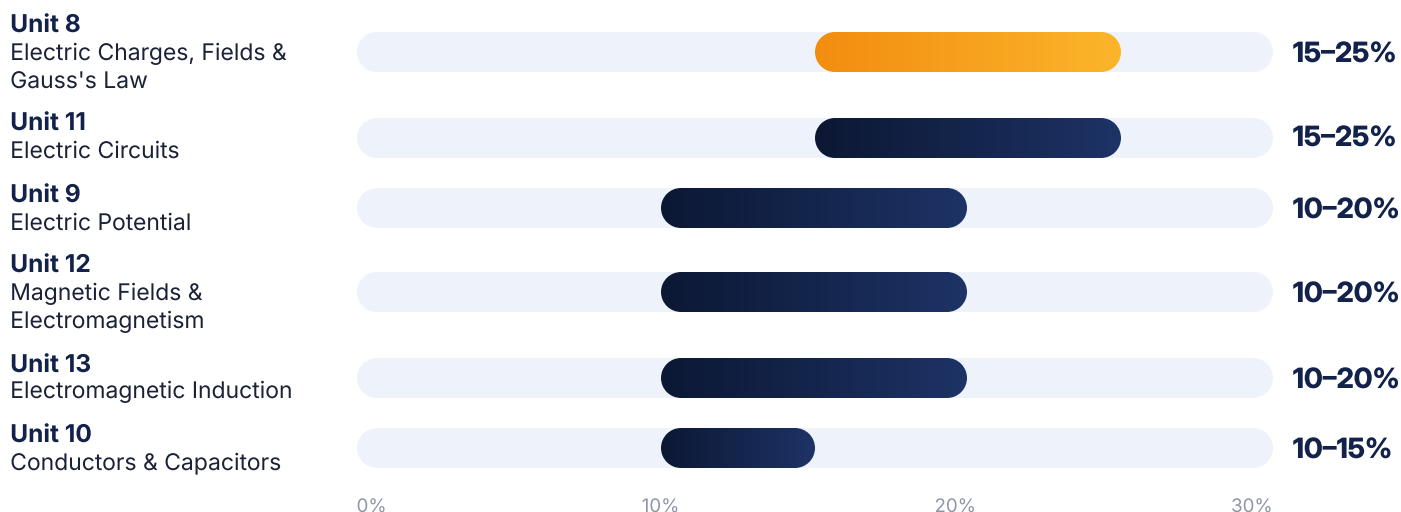
of your total AP score

0

points lost for a wrong answer, so never leave one blank

How the questions are weighted by unit

College Board gives a range for each unit's share of the multiple-choice section. Bars show minimum to maximum.



What this means for your study time: Electric fields with Gauss's law and circuits are the largest units. Magnetism and induction together are a similar share, so don't leave them to April.

Strategies that work on this section

- **Draw a diagram** for every question: forces, fields, circuits or rays. It takes seconds and prevents sign errors.
- **Check limiting cases.** What happens if a mass is zero or an angle is 90° ? Wrong choices often fail this test.
- **Use units to eliminate** answer choices with the wrong dimensions.
- **Read graphs as relationships:** slope and area under the curve both have physical meaning.
- **Work symbolically first**, then plug in numbers only if needed.
- **Flag and move on** after 3 minutes. Return once the quick points are banked.

03 WHAT THE COURSE COVERS

The 6 units, in plain English

Everything on the exam fits into these units.

UNIT

8

15–25%

Electric Charges, Fields & Gauss's Law

Be ready to: apply Coulomb's law; integrate to find fields of continuous charge distributions; use Gauss's law for spheres, cylinders and planes.

UNIT

10

10–15%

Conductors & Capacitors

Be ready to: describe charge on conductors in equilibrium; calculate capacitance; combine capacitors; include dielectrics; find stored energy.

UNIT

12

10–20%

Magnetic Fields & Electromagnetism

Be ready to: find forces on moving charges and currents; use the Biot-Savart law and Ampère's law for wires and solenoids.

UNIT

9

10–20%

Electric Potential

Be ready to: relate field and potential ($E = -dV/dx$); calculate potential of point and continuous charges; find potential energy of charge systems.

UNIT

11

15–25%

Electric Circuits

Be ready to: use Kirchhoff's rules on multi-loop circuits; analyze RC circuits with exponential charging and discharging; calculate power.

UNIT

13

10–20%

Electromagnetic Induction

Be ready to: apply Faraday's and Lenz's laws; analyze motional emf; describe inductance and LR circuits; analyze LC oscillations qualitatively.

Units are numbered 8–13 because they continue on from Physics C: Mechanics (Units 1–7). Mechanics ideas such as forces, energy and Newton's laws are used throughout.

SECTION II

04 The 4 free-response questions

95 minutes, 4 questions. The four question types are the same every year, so you can practice each on purpose.

MR

TBR

LAB

QQT

Mathematical Routines and Lab questions are typically worth the most points

MR

Mathematical Routines

Derive and calculate

Use equations and principles to derive expressions and calculate quantities, often starting symbolically.

Start from a fundamental equation and show each substitution.

TBR

Translation Between Representations

Diagrams, graphs, equations

Draw a free-body diagram, energy bar chart or graph, then write equations that are consistent with it.

Arrows on free-body diagrams start on the object and are labeled.

LAB

Experimental Design & Analysis

Plan and analyze

Design a procedure, choose what to graph to get a straight line (linearize), then use the slope to find a quantity.

Linearize: graph the quantities that make $y = mx + a$ a straight line.

QQT

Qualitative/Quantitative Translation

Explain, then verify

Explain a situation in words with physics reasoning, then derive an equation that agrees with your explanation.

Make sure your words and your equation predict the same trend.

◆ A time plan for the 95 minutes

MR

~25 min

TBR

~22 min

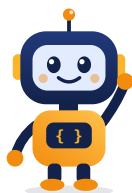
LAB

~25 min

QQT

~23 min

A suggested split, not a rule. Start with the question type you're strongest at, and never leave a question blank.



Points are given for **derivations**. "Plugging in" without a starting equation often loses credit.

05

FREE-RESPONSE PLAYBOOK

Anatomy of a full-credit answer

Here is a short practice prompt with a full-credit answer, and the habits that earn points.

PRACTICE TASK • WRITTEN BY ACT SAT GO

Practice prompt (MR style): A solid nonconducting sphere of radius R has uniform charge density and total charge Q . (a) Use Gauss's law to find the electric field at distance $r < R$ from the center. (b) Find the field for $r > R$. (c) Sketch how E depends on r .

Full-credit response

① (a) Choose a spherical Gaussian surface of radius r . By symmetry E is radial and constant on it: $\oint E \cdot dA = E(4\pi r^2)$.

② Enclosed charge = $Q(r^3/R^3)$. So $E(4\pi r^2) = Qr^3/(\epsilon_0 R^3)$, giving $E = Qr/(4\pi\epsilon_0 R^3)$.

③ (b) For $r > R$, all of Q is enclosed: $E = Q/(4\pi\epsilon_0 r^2)$.

④ (c) E rises linearly from 0 at the center to its maximum at $r = R$, then falls as $1/r^2$.

- ① **Justify the Gaussian surface** with symmetry.
- ② **Find the enclosed charge** using the volume fraction.
- ③ **Apply Gauss's law** separately in each region.
- ④ **Check continuity:** both expressions give the same E at $r = R$.

Five habits for every free-response question

1

Answer every lettered part, in order, with a label.

2

Show the equation, the substitution and the units.

3

Justify with a cause $\rightarrow$ effect sentence, not a word.

4

Use the reference sheet; you don't need to memorize it.

5

Blank part? Write the relationship you would use. It can earn credit.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

The field inside a conductor is **huge!** (It's zero.)



01 Field inside a conductor

In electrostatic equilibrium, $E = 0$ inside a conductor and any excess charge sits on the surface.

02 Gauss's law without symmetry

Gauss's law is always true, but it only gives E easily when symmetry makes E constant on the surface.

03 Sign of $E = -dV/dx$

The field points toward decreasing potential.

04 Capacitor combinations

Capacitors add directly in **parallel** and as reciprocals in **series**, the opposite of resistors.

05 RC behavior at $t = 0$ and ∞

An uncharged capacitor acts like a wire at $t = 0$ and like an open circuit after a long time.

06 Inductor behavior

An inductor resists changes in current: open-circuit-like at $t = 0$, a plain wire after a long time.

07 Right-hand rule for wires

Thumb along current; fingers curl in the direction of B around the wire.

08 Lenz's law direction

The induced current opposes the **change** in flux, not the flux itself.

09 Ampère's law path

Choose a loop where B is constant and parallel to the path, and count only the current enclosed.

10 Plugging in too early

Derive symbolically first. Many questions ask for answers "in terms of" given quantities.

Quick self-check: a charged capacitor discharges through a resistor. After one time constant $\tau = RC$, what fraction of its charge remains? (About 37%, or $1/e$.)

07

PLAN YOUR YEAR

Your study roadmap to May

A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, and keep doing a few problems from older units every week.



- Sep – Oct**
Fields
 - Unit 8** biggest unit
 Coulomb's law, fields of continuous charges by integration, and Gauss's law for every symmetry.
- Nov – Dec**
Potential + capacitors
 - Units 9–10**
 Potential from fields, conductors, capacitance and dielectrics. Start one free-response question per week.
- Jan – Feb**
Circuits
 - Unit 11**
 Kirchhoff's rules and RC circuits with differential equations.
- Mar – Apr**
Magnetism + induction
 - Units 12–13**
 Biot-Savart, Ampère, Faraday, Lenz, inductors and LR circuits. Begin full timed sections in April.
- Early May**
Sharpen
 - Free-response drills**
 All four question types under timing, focusing on Gauss and Ampère derivations.
- Exam day**
May
 - AP Physics C: E&M exam**
 Sleep well, pack your calculator, and use the checklist at the end of this guide.

• The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: 21 multiple-choice questions plus 2 free-response questions	List your two weakest units
3 weeks out	Fix weak units; one of each free-response type	80%+ on unit drills
2 weeks out	Full 3-hour practice exam	Finish both sections on time
Final week	Light review: equation sheet, diagram rules, traps list	Rest the night before

08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Use $g = 9.8 \text{ m/s}^2$ unless stated. Give yourself **22 minutes** for all ten, then check page 10.

1 UNIT 8 • GAUSS'S LAW

A thin spherical shell of radius R carries charge Q spread uniformly. What is the electric field at a distance $R/2$ from the center?

- A $Q/(4\pi\epsilon_0 R^2)$ B $Q/(\pi\epsilon_0 R^2)$
C $Q/(8\pi\epsilon_0 R^2)$ D 0

2 UNIT 8 • SOLID SPHERE

A uniformly charged solid sphere (radius R , charge Q) has field E_0 at its surface. What is the field at $r = R/2$?

- A $E_0/4$ B E_0
C $2E_0$ D $E_0/2$

3 UNIT 9 • FIELD FROM POTENTIAL

The potential in a region is $V(x) = 4x^2 - 3x$ (V, with x in m). What is the electric field at $x = 1$ m?

- A 5 V/m in the $-x$ direction
B 5 V/m in the $+x$ direction
C 1 V/m in the $-x$ direction
D 8 V/m in the $+x$ direction

4 UNIT 10 • CAPACITORS

A parallel-plate capacitor is charged and then disconnected from the battery. The plate separation is doubled. What happens to the stored energy?

- A It halves
B It doubles
C It stays the same
D It quadruples

5 UNIT 11 • RC CIRCUITS

An uncharged $50 \mu\text{F}$ capacitor is charged through a $2.0 \text{ k}\Omega$ resistor by a 10 V battery. After 0.10 s, the capacitor's voltage is closest to:

- A 3.7 V B 5.0 V
C 6.3 V D 10 V

6 UNIT 11 • KIRCHHOFF

Immediately after a switch closes, a 12 V battery is connected to a 6Ω resistor in series with an **uncharged** capacitor. What is the current?

- A 0 A B 1 A
C 2 A D 12 A

7 UNIT 12 • AMPÈRE'S LAW

A long solenoid has 1,000 turns per meter and carries 2.0 A. What is the magnetic field inside? ($\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$)

- A $1.3 \times 10^{-3} \text{ T}$ B $2.5 \times 10^{-3} \text{ T}$
C $5.0 \times 10^{-3} \text{ T}$ D $2.5 \times 10^{-6} \text{ T}$

8 UNIT 12 • LONG WIRE

A long straight wire carries 5.0 A. What is the magnetic field 2.0 cm from the wire?

- A $5.0 \times 10^{-7} \text{ T}$ B $5.0 \times 10^{-5} \text{ T}$
C $1.6 \times 10^{-4} \text{ T}$ D $2.5 \times 10^{-5} \text{ T}$

9 UNIT 13 • FARADAY'S LAW

The flux through a 50-turn coil is $\Phi = 0.02t^2$ (Wb per turn). What is the magnitude of the induced emf at $t = 3$ s?

- A 0.12 V B 0.18 V
C 6.0 V D 9.0 V

10 UNIT 13 • LR CIRCUITS

A 0.50 H inductor and a 10Ω resistor are connected in series to a battery. What is the time constant of the circuit?

- A 0.05 s B 0.5 s
C 5 s D 20 s

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	D • 0	A Gaussian sphere inside the shell encloses no charge, so by symmetry $E = 0$ everywhere inside.
2	D • $E_0/2$	Inside a uniform solid sphere, $E = Qr/(4\pi\epsilon_0 R^3)$ grows linearly with r . Half the radius gives half the surface field.
3	A	$E = -dV/dx = -(8x - 3) = -5 \text{ V/m}$ at $x = 1$, so 5 V/m in the $-x$ direction.
4	B • doubles	Isolated, so Q is constant. Doubling d halves C . $U = Q^2/(2C)$, so U doubles; the extra energy comes from work pulling the plates apart.
5	C • 6.3 V	$\tau = RC = 2,000 \times 50 \times 10^{-6} = 0.10 \text{ s}$. After one time constant, $V = 10(1 - e^{-1}) \approx 6.3 \text{ V}$.
6	C • 2 A	An uncharged capacitor has no voltage across it at $t = 0$, so it acts like a wire: $I = 12/6 = 2 \text{ A}$.
7	B • $2.5 \times 10^{-3} \text{ T}$	$B = \mu_0 nI = (4\pi \times 10^{-7})(1,000)(2.0) \approx 2.5 \times 10^{-3} \text{ T}$.
8	B • $5.0 \times 10^{-5} \text{ T}$	$B = \mu_0 I/(2\pi r) = (2 \times 10^{-7})(5.0)/0.020 = 5.0 \times 10^{-5} \text{ T}$.
9	C • 6.0 V	$\text{emf} = N d\Phi/dt = 50 \times 0.04t = 50 \times 0.12 = 6.0 \text{ V}$ at $t = 3 \text{ s}$. Choice A forgets the 50 turns.
10	A • 0.05 s	$\tau = L/R = 0.50/10 = 0.05 \text{ s}$.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Re-learn Gauss's law and RC circuits (Units 8 and 11) first.

10 KNOW THE TARGET

How students score

In recent years, about 73–75% of students earned a 3 or higher on AP Physics C: Electricity and Magnetism, and about 24–25% earned a 5. Steady preparation is what moves a score up a full point.

73–75%

of students earned a 3 or higher in recent years

24–25%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, plus **two pencils and a dark-ink pen** for the free-response booklet
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ An **approved calculator** with fresh batteries
- ☐ Plan your time: about **2 min** per multiple-choice question and **23 min** per free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ A **ruler or straightedge** for graphs and diagrams, if permitted

In the free-response section: start from a fundamental principle, draw clean labeled diagrams, keep words and equations consistent, and show every step of a derivation before plugging in numbers.



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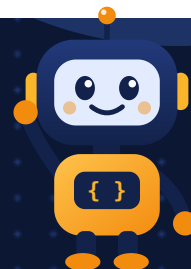
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